

Numerical methods

Problems for the practices

Week 1 – Machine numbers

1.1. List the elements of the set of machine numbers $M = M(3, -2, 2)$!

How many elements does M have?

Find the special machine numbers ε_0 , ε_1 , M_∞ in M !

1.2. Find the floating-point number $\text{fl}(10.85)$ in the set $M(5, -4, 4)$!

1.3. Find the floating-point number $\text{fl}(\frac{1}{6})$ in $M(6, -5, 5)$!

1.4. In machine arithmetic the addition operation is denoted by $\oplus$. Give some examples in the set $M = M(5, -4, 4)$ for the following statements:

(a) $a \oplus b = a$, while $b \neq 0$;

(b) the associative property is not valid.

1.5. In the set of machine numbers $M = M(5, -6, 6)$ find

(a) the machine numbers 1 and 8;

(b) the machine number assigned to 0.12;

(c) the result of the operation $(1 \oplus 8) \oplus \text{fl}(0.12)$!

1.6. In the set of machine numbers $M = M(5, -3, 3)$ find

(a) $\text{fl}(\sqrt{2})$ and $\text{fl}(\sqrt{3})$;

(b) the result of the machine addition $\text{fl}(\sqrt{2}) \oplus \text{fl}(\sqrt{3})$;

(c) the absolute and relative error bounds of the previous values (based on the floating-point representation)!

1.7. In the set of machine numbers $M = M(6, -10, 10)$

$$x_1 = [100000|0] \quad \text{and} \quad x_2 = [111111|-1]$$

are given elements. Complete the machine subtraction $x_1 \ominus x_2$ and find the relative error bounds!